AIVOA PHARMACEUTICALS MANUFACTURING CORP.
GLOBAL QUALITY OPERATIONS & COMPLIANCE DIVISION

# PHARMACEUTICAL API QUALITY DEVIATION & INVESTIGATION REPORT

| | | | |
|---|---|---|---|
| **Deviation Number:** | DEV-2026-1109 | **Classification:** | Critical / Foreign Particulate Contamination |
| **Product / API Name:** | Metformin Hydrochloride API (USP/BP) | **Batch / Lot Number:** | MET-2026-904 |
| **Manufacturing Stage:** | Stage 4: Fluid Bed Drying & Particle Conditioning | **Equipment Tag:** | Fluid Bed Dryer FBD-04 / Sifter SFT-01 |
| **Date of Occurrence:** | 2026-10-18 16:45 UTC | **Date of Discovery:** | 2026-10-19 10:15 UTC |
| **Material / Lot ID:** | In-process Granular Wet Cake LOT-W-881 | **Reporting Unit:** | In-Process Physical Quality Assurance |

## 1. OBSERVED DEVIATION & ANALYTICAL TEST RESULTS

| Test Parameter | Regulatory Specification | Observed Lab Result | Test Methodology / Instrument |
|---|---|---|---|
| Visible Foreign Particulate Matter | **Zero visible foreign particles** | **Metallic micro-filaments detected [OOS]** | Stereomicroscopy & Optical Inspection |
| Loss on Drying (LOD) | NMT 0.50% | 0.22% | Halogen Moisture Analyzer 105°C |
| Assay (Dry Basis) | 98.5% - 101.0% | 99.9% | Potentiometric Titration with HClO4 |
| Related Substance: Dicyandiamide | NMT 0.02% | < 0.005% | HPLC Ion Exchange |

## 2. INCIDENT CHRONOLOGY & FORENSIC NARRATIVE

Routine optical microscopy during sifting of Metformin HCl batch MET-2026-904 detected 7 metallic wire fragments (0.4 mm to 1.2 mm). SEM-EDX elemental analysis confirmed the composition as 316L stainless steel. Visual inspection inside Fluid Bed Dryer FBD-04 discovered mechanical rubbing between the air distribution rotating agitator blade and the lower perforated bottom retention screen.

## 3. IMMEDIATE CONTAINMENT ACTIONS TAKEN

1. Complete batch MET-2026-904 (3,500 kg) quarantined under physical red tag lock.
2. FBD-04 dryer red-tagged; maintenance work order issued for agitator shaft realignment.
3. Preceding and succeeding batches quarantined for 100% magnetic and optical sifter audit.

| | | |
|---|---|---|
| **Prepared by:** | H. Schmidt (QC Physical QA) | 2026-10-19 |
| **Verified by:** | P. O'Connor (Maintenance Lead) | 2026-10-19 |
| **QA Approval:** | Dr. S. Thornton (QA Lead) | **Status: PENDING ROOT CAUSE ANALYSIS** |